Line list and line sizing

Flash drum D-400 unit · process line list with the sizing check in three cases

Document no. FD-V1-PE-LIN-001

Every line of FD-400-PID-001 with its service, flow, density, velocity, pressure drop, design conditions, piping class and insulation, checked against the sizing criteria of the design basis. Lines without a number on the P&ID are numbered here and marked “proposed”.

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Holds and engineering queries

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Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

Rev	Date	Description	Originator	Checked	Approved
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FD-V1-PE-LIN-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

1 · Sizing criteria, classes and insulation

Key	Criterion	v min m/s	v max m/s	ΔP max kPa/100 m	ρv² max Pa
Ld	Liquid, pressure flow (pump discharge, let-down inlet)	0.6	3.0	45.0	—
Lb	Liquid at its bubble point (drum outlet to a control valve)	—	1.0	10.0	—
TP	Two-phase, flashing (control-valve outlet)	—	20.0	30.0	10,000.0
V	Vapour, low-pressure process	—	20.0	10.0	6,000.0
HO	Hot oil, pressure flow	0.3	3.0	40.0	—
N	Nitrogen utility (sized on the regulator capacity)	—	30.0	50.0	—
R	Relief (PSV inlet / outlet, checked in the Relief tab)	—	—	—	—
D	Drain (intermittent, not velocity-sized)	—	—	—	—

ASME B36.10 Sch 40 (STD) for every size (assumed); carbon steel roughness 0.046 mm; Churchill friction factor; liquid viscosity 0.27 − 0.0015·T mPa s (typical light hydrocarbon, assumed); vapour 0.008 mPa s; homogeneous two-phase density.

Class	Description
A1A	Carbon steel, ASME B16.5 class 150, CA 3 mm — process hydrocarbon (assumed from the P&ID class code)
B1A	Carbon steel, ASME B16.5 class 300, CA 1.5 mm — hot oil (assumed from the P&ID class code)
N1A	Carbon steel, class 150, CA 1 mm — nitrogen (assumed; the P&ID shows A1A)
F1A	Carbon steel, class 150, CA 3 mm — flare (assumed)

Insulation	Description
H	heat conservation, mineral wool 50 mm (assumed)
AC	anti-condensation 25 mm (cold feed at 10 °C in humid air, dew point ≈ 25 °C, assumed)
PP	personnel protection only where the surface is > 60 °C within reach (assumed)
—	none

2 · Line list

Line	Seg	NPS	From	To	Service	Stream	Items on the line	Op. P kPa g	Op. T °C	Design P kPa g	Design T °C	MDMT °C	Class	Rating kPa g @ T	Ins.	Criterion	Worst case check	Status
2"-P-4001-A1A		2"	P-400 discharge (battery limit)	E-400 tube inlet (XV-408, FT-400)	Cold feed	2	XV-408, FT-400, TI-400	630.5	10.0	820.0	65.0	0.0	A1A	1,875	AC	Liquid, pressure flow (pump discharge, let-down inlet)	✗ Design, Maximum rate → 3"	P&ID
2"-P-4002-A1A	a	2"	E-400 tube outlet	FCV-400 inlet	Hot feed, liquid under pressure	3	TT-401	610.9	74.0	820.0	205.0	0.0	A1A	1,363	H	Liquid, pressure flow (pump discharge, let-down inlet)	✗ Design, Maximum rate → 3"	P&ID
6"-P-4004-A1A		6"	FCV-400 outlet (HV-401 / HV-402)	D-400 inlet nozzle N1	Let-down outlet, flashing two-phase (drawn on the P&ID as part of 2"-P-4002-A1A)	3A	FCV-400, HV-401, HV-402, HV-403	102.7	49.3	450.0	150.0	0.0	A1A	1,580	—	Two-phase, flashing (control-valve outlet)	✓ all cases	proposed
4"-PV-4021-A1A	a	4"	D-400 top nozzle N2	PCV-420 inlet	Vapour, drum to the pressure-control valve (PSV-400 branch)	4	PT-420, PSHH-426, AI-400, FI-420, AI-420	102.7	49.2	450.0	150.0	0.0	A1A	1,580	—	Vapour, low-pressure process	✓ all cases	P&ID
4"-PV-4021-A1A	b	4"	PCV-420 outlet (HV-421 / HV-422)	Vapour (fuel-gas) header 30 kPa g	Vapour after PCV-420	4	PCV-420, HV-421, HV-422, HV-423	30.0	49.2	450.0	150.0	0.0	A1A	1,580	—	Vapour, low-pressure process	✗ Maximum rate → 6"	P&ID
1"-N-4031-A1A	a	1"	N2 header 700 kPa g	PCV-430 inlet (HV-430)	Nitrogen to the blanket regulator	N1	HV-430	700.0	28.0	1,000.0	65.0	0.0	A1A	1,875	—	Nitrogen utility (sized on the regulator capacity)	✓ all cases	P&ID
1"-N-4031-A1A	b	1"	PCV-430 outlet	D-400 top nozzle N5	Nitrogen blanket into D-400	N1	PCV-430	30.0	28.0	450.0	150.0	0.0	A1A	1,580	—	Nitrogen utility (sized on the regulator capacity)	✓ all cases	P&ID
2"-P-4003-A1A		2"	D-400 bottom nozzle N3	LCV-400 inlet (XV-418)	Drum liquid at its bubble point	5	FT-410, AI-410, TI-410, XV-418	133.6	49.2	450.0	150.0	0.0	A1A	1,580	—	Liquid at its bubble point (drum outlet to a control valve)	✗ Design, Maximum rate, Turndown → 4"	P&ID
4"-P-4005-A1A		4"	LCV-400 outlet (HV-411 / HV-412)	Downstream receiver 40 kPa g	Bottoms after LCV-400, flashing (drawn as part of 2"-P-4003-A1A)	5A	LCV-400, HV-411, HV-412, HV-413	55.7	42.3	450.0	150.0	0.0	A1A	1,580	—	Two-phase, flashing (control-valve outlet)	✓ all cases	proposed
2"-HO-4011-B1A		2"	Hot-oil supply header	TCV-401 inlet (XV-409)	Hot-oil supply, total 3-way flow	H1	XV-409, TI-402, PDI-406	500.0	180.0	1,000.0	205.0	0.0	B1A	4,361	H	Hot oil, pressure flow	✗ Design, Maximum rate, Turndown → 4"	P&ID
3"-HO-4013-B1A		3"	TCV-401 (E-400 port)	E-400 shell inlet	Hot oil to E-400 (not numbered on the P&ID)	H2	FT-405	480.0	180.0	1,000.0	205.0	0.0	B1A	4,361	H	Hot oil, pressure flow	✓ all cases	proposed
3"-HO-4015-B1A		3"	E-400 shell outlet	Return junction	Hot oil from E-400 (not numbered on the P&ID)	H3	TI-403	370.0	124.0	1,000.0	205.0	0.0	B1A	4,361	H	Hot oil, pressure flow	✓ all cases	proposed
4"-HO-4014-B1A		4"	TCV-401 (bypass port)	Return junction	Hot-oil bypass of the 3-way valve (not numbered on the P&ID)	H4	—	370.0	180.0	1,000.0	205.0	0.0	B1A	4,361	H	Hot oil, pressure flow	✓ all cases	proposed
2"-HO-4012-B1A		2"	Return junction	Hot-oil return header	Hot-oil return, total 3-way flow	H5	—	350.0	168.6	1,000.0	205.0	0.0	B1A	4,361	H	Hot oil, pressure flow	✗ Design, Maximum rate, Turndown → 4"	P&ID
3"-PV-4023-A1A		3"	4"-PV-4021-A1A (D-400 top)	PSV-400 inlet	PSV-400 inlet (not numbered on the P&ID)	F1	PSV-400 inlet block (CSO, assumed)	—	—	450.0	150.0	0.0	A1A	1,580	—	Relief (PSV inlet / outlet, checked in the Relief tab)	see Relief tab	proposed
6"-FL-4041-F1A		6"	PSV-400 outlet	Flare header (pipe rack)	PSV-400 tail pipe (not numbered on the P&ID)	F1	PSV-400 outlet block (CSO, assumed)	—	—	350.0	150.0	0.0	F1A	1,580	—	Relief (PSV inlet / outlet, checked in the Relief tab)	see Relief tab	proposed
2"-CD-4051-A1A		2"	D-400 bottom (HV-415)	Closed drain	Drum drain, intermittent (not numbered on the P&ID)	—	HV-415	—	—	450.0	150.0	0.0	A1A	1,580	—	Drain (intermittent, not velocity-sized)	intermittent	proposed

3 · Sizing in the three cases

Line	Case	kg/h	ρ kg/m³	μ mPa s	T °C	P kPa g	m³/h	v m/s	ρv² Pa	Re	f	ΔP kPa/100 m	Check	Smallest size	Note
2"-P-4001-A1A	Design	10,208	638.01	0.255	10.0	600.0	16.0	2.05	2,689	269,685	0.0203	52.05	✗ ΔP > 45	3"	
2"-P-4001-A1A	Maximum rate	12,760	638.01	0.255	10.0	571.9	20.0	2.57	4,202	337,106	0.0201	80.44	✗ ΔP > 45	3"	
2"-P-4001-A1A	Turndown	6,380	638.01	0.255	10.0	630.5	10.0	1.28	1,051	168,553	0.0209	20.95	✓	2"	
2"-P-4002-A1A (a)	Design	10,208	571.81	0.159	74.0	550.0	17.9	2.29	3,001	432,513	0.0199	56.85	✗ ΔP > 45	3"	
2"-P-4002-A1A (a)	Maximum rate	12,760	571.81	0.159	74.0	493.8	22.3	2.86	4,689	540,641	0.0197	88.15	✗ ΔP > 45	3"	
2"-P-4002-A1A (a)	Turndown	6,380	571.81	0.159	74.0	610.9	11.2	1.43	1,172	270,321	0.0203	22.68	✓	2"	
6"-P-4004-A1A	Design	10,208	27.12	0.196	49.3	102.7	376.4	5.61	852	119,549	0.0190	5.25	✓	6"	homogeneous two-phase, 17 wt% vapour
6"-P-4004-A1A	Maximum rate	12,760	27.12	0.196	49.3	102.7	470.5	7.01	1,332	149,436	0.0184	7.96	✓	6"	homogeneous two-phase, 17 wt% vapour
6"-P-4004-A1A	Turndown	6,380	27.12	0.196	49.3	102.7	235.3	3.50	333	74,718	0.0204	2.21	✓	4"	homogeneous two-phase, 17 wt% vapour
4"-PV-4021-A1A (a)	Design	1,714	4.79	0.008	49.2	102.7	358.1	12.10	701	740,567	0.0171	5.87	✓	4"	
4"-PV-4021-A1A (a)	Maximum rate	2,147	4.79	0.008	49.2	102.7	448.8	15.17	1,101	927,952	0.0170	9.14	✓	4"	
4"-PV-4021-A1A (a)	Turndown	1,063	4.79	0.008	49.2	102.7	222.2	7.51	270	459,486	0.0175	2.31	✓	3"	
4"-PV-4021-A1A (b)	Design	1,714	3.08	0.008	49.2	30.0	556.3	18.80	1,089	740,567	0.0171	9.12	✓	4"	
4"-PV-4021-A1A (b)	Maximum rate	2,147	3.08	0.008	49.2	30.0	697.2	23.56	1,710	927,952	0.0170	14.20	✗ v > 20, ΔP > 10	6"	
4"-PV-4021-A1A (b)	Turndown	1,063	3.08	0.008	49.2	30.0	345.1	11.66	419	459,486	0.0175	3.59	✓	4"	
1"-N-4031-A1A (a)	Design	21	8.97	0.018	28.0	700.0	2.3	1.15	12	15,195	0.0311	0.69	✓	0.75"	blanket demand: drum emptied at 14 m³/h with no feed (normally no flow)
1"-N-4031-A1A (a)	Maximum rate	21	8.97	0.018	28.0	700.0	2.3	1.15	12	15,195	0.0311	0.69	✓	0.75"	blanket demand: drum emptied at 14 m³/h with no feed (normally no flow)
1"-N-4031-A1A (a)	Turndown	21	8.97	0.018	28.0	700.0	2.3	1.15	12	15,195	0.0311	0.69	✓	0.75"	blanket demand: drum emptied at 14 m³/h with no feed (normally no flow)
1"-N-4031-A1A (b)	Design	21	1.47	0.018	28.0	30.0	14.0	7.00	72	15,195	0.0311	4.20	✓	0.75"	blanket demand: drum emptied at 14 m³/h with no feed (normally no flow)
1"-N-4031-A1A (b)	Maximum rate	21	1.47	0.018	28.0	30.0	14.0	7.00	72	15,195	0.0311	4.20	✓	0.75"	blanket demand: drum emptied at 14 m³/h with no feed (normally no flow)
1"-N-4031-A1A (b)	Turndown	21	1.47	0.018	28.0	30.0	14.0	7.00	72	15,195	0.0311	4.20	✓	0.75"	blanket demand: drum emptied at 14 m³/h with no feed (normally no flow)
2"-P-4003-A1A	Design	8,494	605.41	0.196	49.2	133.6	14.0	1.80	1,962	291,719	0.0202	37.83	✗ v > 1, ΔP > 10	3"	
2"-P-4003-A1A	Maximum rate	10,613	605.41	0.196	49.2	133.6	17.5	2.25	3,064	364,543	0.0200	58.44	✗ v > 1, ΔP > 10	4"	
2"-P-4003-A1A	Turndown	5,320	605.41	0.196	49.2	133.6	8.8	1.13	770	182,599	0.0208	15.26	✗ v > 1, ΔP > 10	3"	
4"-P-4005-A1A	Design	8,494	62.77	0.208	41.4	50.0	135.3	4.57	1,313	141,212	0.0194	12.42	✓	4"	homogeneous two-phase, 5.3 wt% vapour
4"-P-4005-A1A	Maximum rate	10,613	72.17	0.207	42.3	55.7	147.1	4.97	1,783	177,661	0.0189	16.45	✓	4"	homogeneous two-phase, 4.7 wt% vapour
4"-P-4005-A1A	Turndown	5,320	54.37	0.210	40.2	43.9	97.8	3.31	594	87,732	0.0206	5.97	✓	3"	homogeneous two-phase, 5.9 wt% vapour
2"-HO-4011-B1A	Design	36,000	800.00	1.000	180.0	500.0	45.0	5.77	26,674	242,522	0.0204	519.33	✗ v > 3, ΔP > 40	4"	
2"-HO-4011-B1A	Maximum rate	36,000	800.00	1.000	180.0	500.0	45.0	5.77	26,674	242,522	0.0204	519.33	✗ v > 3, ΔP > 40	4"	
2"-HO-4011-B1A	Turndown	36,000	800.00	1.000	180.0	500.0	45.0	5.77	26,674	242,522	0.0204	519.33	✗ v > 3, ΔP > 40	4"	
3"-HO-4013-B1A	Design	10,980	800.00	1.000	180.0	480.0	13.7	0.80	512	49,851	0.0229	7.54	✓	3"	
3"-HO-4013-B1A	Maximum rate	14,692	800.00	1.000	180.0	480.0	18.4	1.07	917	66,705	0.0219	12.91	✓	3"	
3"-HO-4013-B1A	Turndown	6,015	800.00	1.000	180.0	480.0	7.5	0.44	154	27,308	0.0256	2.52	✓	2"	
3"-HO-4015-B1A	Design	10,980	800.00	1.000	120.1	370.0	13.7	0.80	512	49,851	0.0229	7.54	✓	3"	
3"-HO-4015-B1A	Maximum rate	14,692	800.00	1.000	124.0	370.0	18.4	1.07	917	66,705	0.0219	12.91	✓	3"	
3"-HO-4015-B1A	Turndown	6,015	800.00	1.000	111.7	370.0	7.5	0.44	154	27,308	0.0256	2.52	✓	2"	
4"-HO-4014-B1A	Design	25,020	800.00	1.000	180.0	370.0	31.3	1.06	894	86,501	0.0206	9.00	✓	3"	
4"-HO-4014-B1A	Maximum rate	21,308	800.00	1.000	180.0	370.0	26.6	0.90	648	73,666	0.0211	6.68	✓	3"	
4"-HO-4014-B1A	Turndown	29,985	800.00	1.000	180.0	370.0	37.5	1.27	1,284	103,666	0.0201	12.61	✓	4"	
2"-HO-4012-B1A	Design	36,000	800.00	1.000	161.7	350.0	45.0	5.77	26,674	242,522	0.0204	519.33	✗ v > 3, ΔP > 40	4"	
2"-HO-4012-B1A	Maximum rate	36,000	800.00	1.000	157.2	350.0	45.0	5.77	26,674	242,522	0.0204	519.33	✗ v > 3, ΔP > 40	4"	
2"-HO-4012-B1A	Turndown	36,000	800.00	1.000	168.6	350.0	45.0	5.77	26,674	242,522	0.0204	519.33	✗ v > 3, ΔP > 40	4"	
3"-PV-4023-A1A	—	—	—	—	—	—	—	—	—	—	—	—	—	—	sized in the Relief tab (3 % inlet loss / built-up back-pressure)
6"-FL-4041-F1A	—	—	—	—	—	—	—	—	—	—	—	—	—	—	sized in the Relief tab (3 % inlet loss / built-up back-pressure)
2"-CD-4051-A1A	—	—	—	—	—	—	—	—	—	—	—	—	—	—	intermittent; size by the drain-out time

6"-P-4004-A1A drawn on the P&ID as 2": Design v 48.3 m/s, ρv² 63,270 Pa, ✗ v > 20, ΔP > 30, ρv² > 10000; Maximum rate v 60.4 m/s, ρv² 98,860 Pa, ✗ v > 20, ΔP > 30, ρv² > 10000; Turndown v 30.2 m/s, ρv² 24,715 Pa, ✗ v > 20, ΔP > 30, ρv² > 10000
4"-P-4005-A1A drawn on the P&ID as 2": Design v 17.4 m/s, ρv² 18,927 Pa, ✗ ΔP > 30, ρv² > 10000; Maximum rate v 18.9 m/s, ρv² 25,700 Pa, ✗ ΔP > 30, ρv² > 10000; Turndown v 12.6 m/s, ρv² 8,569 Pa, ✗ ΔP > 30

4 · Engineering queries

No.	Subject	Query	Why it matters	Proposed answer	Status
EP-01	Let-down outlet drawn 2"	FD-400-PID-001 draws the line from FCV-400 to D-400 as part of 2"-P-4002-A1A. Downstream of FCV-400 the feed flashes (17 wt% vapour at 102.7 kPa g): in 2" the velocity would be 48.3 m/s and ρv² 63270 Pa at design (60.38 m/s at the maximum rate).	Two-phase flow at that velocity erodes the pipe, vibrates and makes slugs; the drum inlet nozzle is 6" in the Design tab.	Give the let-down outlet its own number at the spec break (FCV-400): 6"-P-4004-A1A (smallest size inside the two-phase criteria at the maximum rate), FCV-400 placed close to D-400, nozzle N1 6".	OPEN
EP-03	Bottoms line 2"-P-4003-A1A at the bubble point	D-400 liquid leaves at its bubble point. In 2" it runs at 1.8 m/s (2.25 m/s at the maximum rate) against ≤ 1.0 m/s for bubble-point liquid; the Design tab sizes the outlet nozzle 3".	Friction and acceleration at the nozzle make vapour before LCV-400 (cavitation, unstable level control, FT-410 error).	Make the line 4" from N3 to LCV-400 (XV-418 and FT-410 with it).	OPEN
EP-05	Hot-oil lines carry the full 3-way flow	TCV-401 is a 3-way valve: 2"-HO-4011-B1A and 2"-HO-4012-B1A carry the whole 36 t/h, not only the 11 t/h to E-400. In 2" that is 5.77 m/s and 519.33 kPa/100 m.	The header ΔP of 150 kPa (PDI-406) would be spent in the piping instead of the valve and E-400.	Make the supply and return 4"; the E-400 branches 3"-HO-4013-B1A / 3"-HO-4015-B1A and the bypass 4"-HO-4014-B1A follow the criteria (3").	OPEN
EP-07	Feed system design pressure not stated	The P&ID gives the supply as 600 kPa g at design; the model's P-400 curve shuts off at 650 kPa g. Rule (1) on the shut-off gives 820 kPa g for the feed line, E-400 tubes and FCV-400 (class 150 is enough).	Every item upstream of FCV-400 must hold the pump shut-off; the D-400 side is protected by PSV-400.	State P-400 maximum shut-off (maximum suction + maximum differential) in the design basis; spec break at FCV-400: 820 / 450 kPa g.	OPEN
EP-10	2" feed lines above the ΔP criterion	2"-P-4001-A1A and the hot-feed segment of 2"-P-4002-A1A run at 52.05 / 56.85 kPa/100 m at design (80.44 / 88.15 at the maximum rate) against 45. The model's lumped line loss is 10 kPa at design, i.e. about 19 m of 2" pipe.	Line friction eats into the 191 kPa margin between the FCV-400 inlet pressure and the bubble pressure of the hot feed.	Keep 2" only if the routed length stays below ≈ 19 m (Layout tab); otherwise 3".	OPEN
EP-11	Vapour line after PCV-420	Downstream of PCV-420 the vapour (4"-PV-4021-A1A, segment b) expands to the 30 kPa g header: in 4" it runs at 18.8 m/s (23.56 m/s and 14.2 kPa/100 m at the maximum rate).	The header back-pressure plus line loss sets how much ΔP is left for PCV-420 (Operating windows tab: the header limits the vapour valve).	Expand to 6" after PCV-420 (the usual expansion after a gas control valve).	OPEN